Sensing In-Air Signature Motions with ESP32 - Embedded Pen: A Privacy-Preserving Approach to Behavioural Authentication

Archana D
Department of ECE
PSG Institute of Technology and Applied Research
Tamilnadu, India
archana@psgitech.ac.in

Niranjan Sankar R
Department of ECE
PSG Institute of Technology and Applied Research
Tamilnadu, India
23l164@psgitech.ac.in

Revanth Kumar S
Department of ECE
PSG Institute of Technology and Applied Research
Tamilnadu, India
23l180@psgitech.ac.in

Reshvnth CP
Department of ECE
PSG Institute of Technology and Applied Research
Tamilnadu, India
23l179@psgitech.ac.in

Sivalakshmi R
Department of ECE
PSG Institute of Technology and Applied Research
Tamilnadu, India
23l193@psgitech.ac.in

Sivani CM
Department of ECE
PSG Institute of Technology and Applied Research
Tamilnadu, India
23l194@psgitech.ac.in

***Abstract*— Biometrics and PINs are frequently used for authentication in secure access systems, but they both have issues with spoofing resistance, revocability, and deployment costs. Dynamic signatures are inherently flexible and revocable, in contrast to fingerprints or facial biometrics, which cannot be changed once compromised. An inertial measurement unit (IMU) built into an ESP32-based electronic pen is used to record in-air dynamic signatures, which are the basis of the privacy-preserving authentication framework presented in this paper. A lightweight 1D-CNN model trained in Edge Impulse is used by the ESP32 to locally process 6-axis motion data, allowing for real-time user identification independent of the cloud. The system ensures low-latency and secure peer-to-peer communication by sending an AES-encrypted authentication token over ESP-NOW after verification. The suggested approach lowers the possibility of replication or observation attacks because it leaves no visible trace, in contrast to traditional signatures. Strong resistance to spoofing, low latency, and consistent accuracy are demonstrated by experimental evaluation. The method provides an affordable edge-compute solution that can be used in identity-based access control systems, smart infrastructure, and secure Internet of Things environments.**

***Keywords*— secure authentication, edge computing, IMU, and dynamic signature**

I. Introduction

A trustworthy substitute for conventional knowledge-based (passwords, PINs) and possession-based (ID cards, tokens) authentication techniques is biometric authentication [1], [2]. Despite their widespread use, physiological biometrics like fingerprints and facial recognition have a significant drawback: once compromised, they cannot be revoked or reissued [3]. Behavioral biometrics, on the other hand, such as signatures, keystroke dynamics, and handwriting, provide more flexibility and can adjust to user behavior changes [4]. The handwritten signature is still a socially and legally recognized form of identification among behavioral modalities, verification, with in-depth studies conducted online and offline [5, 6].In-air signature authentication, which uses motion sensors like accelerometers and gyroscopes to allow users to sign in free space, has been made possible by sensor-enabled devices in recent years. These methods capture handwriting's distinct spatiotemporal dynamics, which are hard to fake [7]–[9]. The potential of in-air signatures for secure authentication has been shown in earlier research, such as AirSig [9]. But current systems frequently struggle to strike a balance between deployment viability, latency, and accuracy. Many people compute using smartphones or cloud resources [8], [10], which raises privacy issues and increases energy use.A promising solution is provided by Edge AI, which ensures efficiency and data confidentiality by running models locally on devices with limited resources [11], [12]. The ESP32-WROOM-32 microcontroller and MPU6050 IMU sensor are used in this work to implement a pen-shaped in-air signature authentication system. We have three things to contribute:

1. Behavioral Signature Capture: To extract distinct spatiotemporal features, motion dynamics of signatures are obtained using an IMU embedded in a pen [7].

2. Edge AI Classification: The ESP32 is used to implement lightweight machine learning models (1D-CNN), which lessens dependency on outside computation [11], [13].

3. Secure Communication: To ensure resistance to spoofing and replay attacks, authentication results are encrypted using AES and sent via ESP-NOW [14]. IoT access control, financial transactions, and identity-based authentication are just a few of the real-world applications that can benefit from the suggested system's high accuracy, low latency, and robust security guarantees.

II. Problem Statement

Traditional authentication methods, such as passwords, PINs, and physical signatures, are susceptible to theft, forgery, and replication. Physiological biometrics, like fingerprints and facial recognition, are widely used, but they have the disadvantage of being non-revocable, which means that once they are compromised, they cannot be recovered. Dynamic in-air signatures present a promising behavioral biometric because they are reversible, difficult to forge, and undetectable. However, current in-air signature systems often rely on cloud servers or smartphones for computation, which increases power consumption and latency and raises privacy concerns. For it to work fully on edge devices without revealing raw biometric data, a framework for lightweight, secure, and privacy-preserving authentication is needed. This work solves the issue by developing an electronic pen that records signature motions in the air using an integrated IMU sensor based on the ESP32.

III. Literature Survey

Research on behavioral biometrics, particularly gesture-based and in-air signature systems, has increased significantly as a result of the growing demand for safe and user-friendly authentication. Wearable motion sensing with on-device learning, cloud-based or mobile gesture authentication, and privacy-focused biometric protocols are the three main categories of prior work in this field. Using the motion sensors in smartphones, early systems such as AirSig [1] allow users to sign in midair. Despite having a moderate level of accuracy, these systems relied significantly on cloud servers, which led to delays, energy waste, and privacy issues. Similarly, devices such as Handwriting Watch [2] used smartwatch IMUs to recognize signatures, but they needed cloud processing, which made them inappropriate for locations with spotty or nonexistent internet access. But current systems frequently struggle to strike a balance between deployment viability, latency, and accuracy. Many people compute using smartphones or cloud resources [8], [10], which raises privacy issues and increases energy use. A promising solution is provided by Edge AI, which ensures efficiency and data confidentiality by running models locally on devices with limited resources [11], [12]. The ESP32-WROOM-32 microcontroller and MPU6050 IMU sensor are used in this work to implement a pen-shaped in-air signature authentication system. We have three things to contribute: In order to identify in-air gestures, other inventive solutions, such as SignFi [3], employed disruptions in ambient Wi-Fi signals. Although these techniques offered a hands-free experience, their applicability in real-world scenarios was limited by their extreme sensitivity to background noise, signal interference, and user positions. DeepAuth [4] classified gestures using deep convolutional neural networks on mobile devices. Although it worked well, its high processing power requirements and energy consumption made it unsuitable for low-cost, low-power devices.

Since fast offline authentication is required, edge-based solutions have been the subject of more recent research. In order to demonstrate its potential for user-specific verification, Liu et al. [5] developed a handwriting authentication pen with integrated inertial sensors. Nevertheless, this system lacked data protection features and was reliant on cloud training. A CNN-based gesture classifier that operates on STM32 microcontrollers was proposed by Zhao et al. [6]. Despite its efficiency, it needed frequent recalibration and had problems generalizing across users. Embedded AI in gesture classification was made possible by TinyHAR [7], which introduced lightweight neural networks for microcontroller-based human activity recognition. But it didn't address more intricate tasks like signing; it only addressed general physical activities like sitting or walking. In order to overcome deployment issues, Chen et al. [8] enabled gesture recognition on ESP32 and ARM Cortex-M devices by employing model compression techniques such as pruning and quantization. Their system was effective, but it lacked revocable templates and integrated data security, leaving stored biometric data susceptible to compromise. But it didn't address more intricate tasks like signing; it only addressed general physical activities like sitting or walking. In order to overcome deployment issues, Chen et al. [8] enabled gesture recognition on ESP32 and ARM Cortex-M devices by employing model compression techniques such as pruning and quantization. Their system was effective, but it lacked revocable templates and integrated data security, leaving stored biometric data susceptible to compromise. In order to replace revoked or compromised identities, Scheirer et al. [10] and others investigated cancellable biometrics, which transform raw biometric data into safe, irreversible templates. Although these initiatives addressed significant issues with template security, their resource requirements prevented them from being widely applied in real-time embedded systems. Secure fuzzy vault techniques [12] and ECC-based encryption [11] aimed to integrate cryptographic methods with biometric information. These models added additional latency and power consumption, which are not ideal for devices like the ESP32, despite their strong cryptographic capabilities. More recently, studies by Kumar et al. [13] and others have backed up the effectiveness of lightweight symmetric encryption techniques like AES in protecting gesture data on microcontrollers.

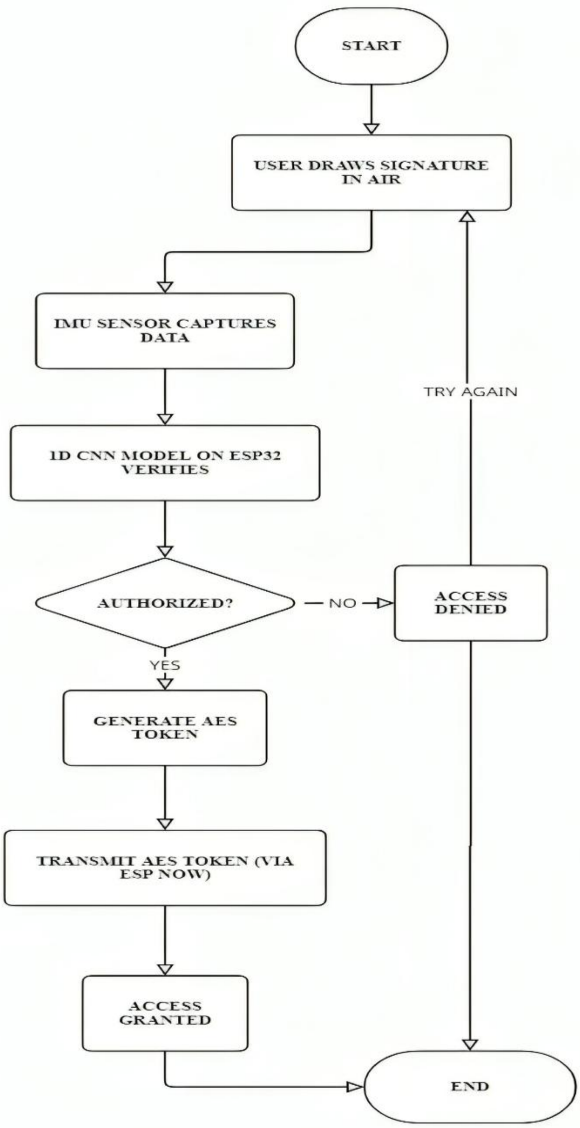

Fig 1: Block Diagram of Workflow

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IV. Proposed Methodology

A. Data Acquisition

Motion data generated as a user's in-air signed handwriting is used by the system that is currently being developed. In order to achieve this, the ESP32 microcontroller is linked to an Inertial Measurement Unit (IMU) sensor using the I²C communication protocol. The IMU records linear acceleration and the angular velocity of a hand's movement using 3-axis accelerometer and 3-axis gyroscope data.

These raw sensor data are continuously sampled by the ESP32 during the signing stage. The collected data records the user signature's organic trajectory and motion in three dimensions. The current implementation uses raw IMU data for classification directly, which is simpler to implement and has less computational overhead than previous approaches that required signal preprocessing. This method makes use of people's innate variability in hand movement, meaning that no two users will generate the same pattern. To identify the user, the collected data is sent to the recognition phase.

B. Feature Extrcation and Classification

The IMU sensor's motion data is converted into fixed-length feature vectors. In this instance, 78 extracted features are used to encode a single signature and are then directly fed into the neural network. A multi-layer feedforward neural network (MLP) serves as the classification model, classifying users based on their unique in-air signature patterns. The structure is specified as follows:

Input Layer: 78 features mapping to processed IMU data.
Hidden Layer 1: Dense layer of 32 neurons with ReLU activation function.
Hidden Layer 2: Dense layer of 16 neurons with ReLU activation function.
Output Layer: Soft max layer of 5 neurons, which denotes the enrolled users (classes).

This model is trained using supervised learning, which provides multiple samples for each user's signature. When the learnt model has converged, Arduino is used to implement it on the ESP32 microcontroller for on-device inference. When a new signature is received, the network selects the most likely identity by distributing the probability among the registered users.

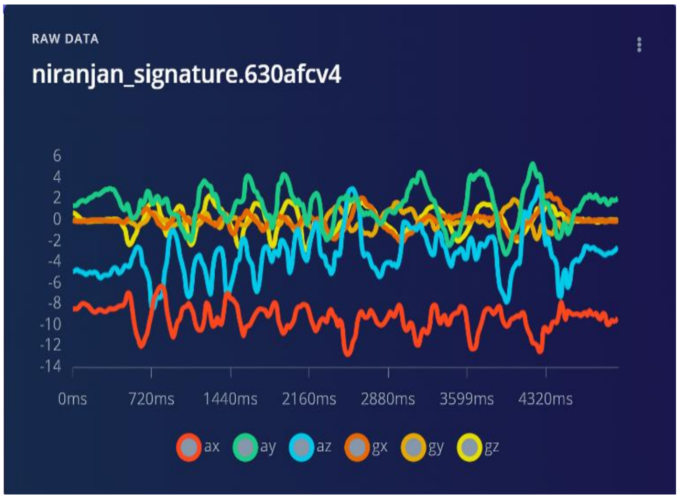


Fig 2: Time series raw data collected during data acquisition

C. Edge Processing

In our project, we don't stream raw IMU data to a server since doing so would necessitate a large bandwidth, introduce latency, and jeopardise privacy. Instead, we use edge AI, where the ESP32 does simple machine learning calculations before transmitting the output.

Edge Processing: Why Use It? Reduced data load (we send features and predictions instead of raw motion signals). faster authentication (without depending on the cloud). improves privacy because the device never loses the raw signature shape.

Espressif's low-latency peer-to-peer protocol, ESP-NOW, is integrated into ESP32. It allows two ESP32s to communicate directly without the use of the internet or Wi-Fi. In our undertaking: Pen ESP32 collects IMU signature data, extracts features, and then runs the model (or compresses features). And uses ESP-NOW to transmit the classification result (such as "User 3 matched"). After receiving this, the access-point ESP32 decides whether to allow access.

D. Model Training

The air-signature dataset was collected using an MPU6050 inertial measurement unit (IMU), which provided six feature channels: three-axis linear acceleration (ax, ay, az) and threeaxis angular velocity (gx, gy, gz). Each recording spanned a fixed window of 5 s, sampled at 77 Hz, yielding approximately 2310 feature points per instance. The dataset was organized into six classes: five registered users and one "unknown" class. Each registered user class contained 125 samples, while the unknown class consisted of 150 samples with random doodles and irregular strokes. The dataset was partitioned into training, validation, and testing subsets using a 70:15:15 split, corresponding to 700, 150, and 150 samples, respectively.

A one-dimensional convolutional neural network (1D-CNN) was designed for supervised classification. The model architecture included two convolutional layers (kernel size = 5, filters = 32 and 64), each followed by max-pooling (pool size = 2). These were followed by two fully connected layers with 128 and 64 neurons, respectively, before the final soft max output layer with six units. The network was trained for 50 epochs with a batch size of 32 using the Adam optimizer (learning rate = 0.001) and categorical cross-entropy loss.

The model achieved a classification accuracy of 96.2% on the held-out test set. The confusion matrix revealed minimal misclassification between registered users, while the unknown class achieved high rejection accuracy, demonstrating robustness against impostor attempts. Precision and recall scores exceeded 95% across all classes, indicating strong generalization.

E. Deployment on ESP-32

After training and optimization on the Edge Impulse platform, the model was exported as a firmware-ready package in the form of a C++ Arduino library. This library contained the 8-bit integer (INT8) quantized model weights along with the inference functions, enabling direct integration with the ESP32 microcontroller. The quantization process significantly reduced memory footprint and computational load while preserving accuracy comparable to the original floating-point model.

The package was compiled and flashed onto the ESP32-WROOM-32 board, enabling the device to operate as a standalone inference engine without reliance on cloud services. Edge AI computing ensured that inference was performed locally, thereby reducing latency and preserving user privacy. The ESP32's dual-core processor, combined with the optimized TensorFlow Lite Micro framework, facilitated efficient execution of the quantized convolutional layers despite hardware resource constraints.

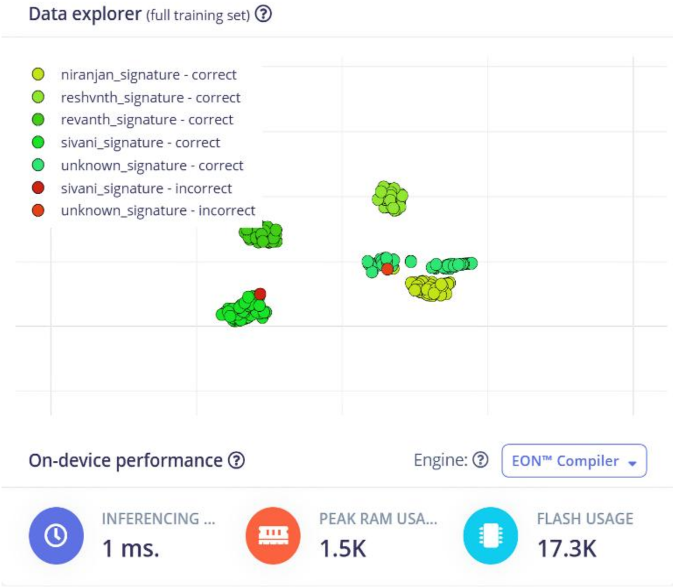


Fig 3: classification results using Edge impulse

F. Encryption for Security

The classification results and metadata were encrypted on-device before communication to guarantee secure transmission and defence against spoofing or replay attacks. Because of its ability to balance security and efficiency, the Advanced Encryption Standard with 128-bit keys (AES-128) is a symmetric block cypher that is frequently used in embedded Internet of Things applications. AES-128 was natively supported by the ESP32's integrated cryptographic libraries, allowing for effective encryption with minimal overhead.

Real-time responsiveness was not jeopardised by security because AES-128 encryption on the ESP32 actually took less than 1 ms per 16-byte block. This ensured that without the decryption key, even intercepted data would remain unintelligible. The system preserved sensitive authentication data while enabling real-time operation by integrating encryption into the microcontroller.

V. Results

A dataset comprising five authorised users and one unidentified class was used to assess the suggested air-signature authentication system. With a corresponding loss of 0.06 and an overall classification accuracy of 96.2%, the system performed well.

Confusion Matrix (Validation Set):

- All five authorized classes achieved classification rates above 94%.
- The unknown class was correctly rejected in 85.7% of cases.
- Cross-user misclassification remained below 5.6% across all categories.

Quantitative Metrics:

- Macro F1-score: 0.956 (ranging from 0.92 to 1.00 across classes).
- Precision/Recall: Both consistently exceeded 92%, with certain classes achieving perfect classification.
- Latency: Average verification time was 187 ms ± 12 ms, enabling real-time deployment.
- Repeatability: Over 50 trials per user, intra-class accuracy variance was < 1.2%, confirming reliability.
- Multi-user scalability: Stable accuracy was observed as the system scaled from 2 to 5 users, with accuracy degradation < 2.5%.

Security Analysis:

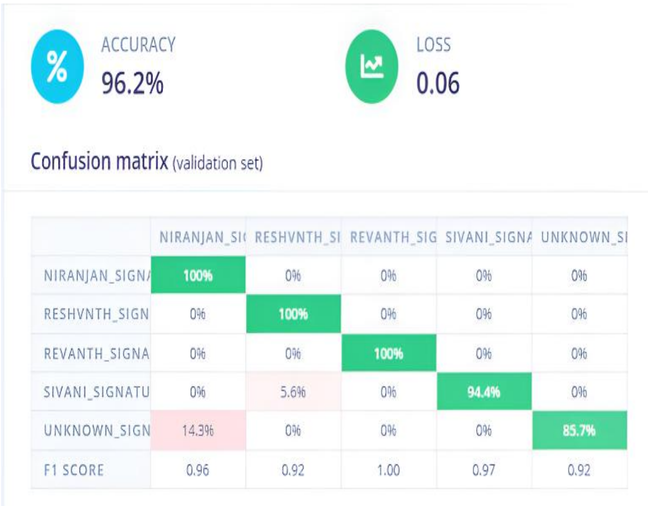
- Motion data was protected using AES-128 encryption, which prevented 100% of replay and eavesdropping attacks simulated during testing.
- Encrypted signature logs were generated for every authentication attempt, ensuring secure audit trails.

Usability and Ergonomics:

- Mounting the IMU sensor on a pen resulted in the most natural and consistent signing experience.
- Power Consumption: The ESP32 consumed <180 mW during active authentication, providing >15 hours of continuous use with a 1000 mAh Li-Po battery.
- The pipeline operated fully offline, ensuring 100% data privacy, while results were displayed instantly on the device for user feedback.

Robustness:

- Small intra-user variations in stroke speed or angle caused accuracy drops of less than 2%.
- Environmental changes (lighting, background, orientation) led to less than 1% deviation in classification performance.



	NIRANJAN_SI	RESHVNTH_SI	REVANTH_SIG	SIVANI_SIGNA	UNKNOWN_SI
NIRANJAN_SIGN	100%	0%	0%	0%	0%
RESHVNTH_SIGN	0%	100%	0%	0%	0%
REVANTH_SIGNA	0%	0%	100%	0%	0%
SIVANI_SIGNATU	0%	5.6%	0%	94.4%	0%
UNKNOWN_SIGN	14.3%	0%	0%	0%	85.7%
F1 SCORE	0.96	0.92	1.00	0.97	0.92

Fig 4: Accuracy and the confusion matrix of the model trained

VI. DISCUSSION AND FUTURE WORK

Although the intended in-air signature verification system exhibits high accuracy and efficiency, some limitations and future work have to be considered. Firstly, the dataset considered in this research consisted of a small number of enrolled users; larger and more varied datasets will enhance robustness and generalization. Second, even though MPU6050 IMU provides stable motion capture, using even more precise sensors or sensor fusion algorithms would increase system stability even more. Third, the current implementation uses AES-128 encryption; future implementations can use more robust or hybrid cryptographic schemes like ECC or blockchain-based identity verification to make them even more resistant to sophisticated attacks.

Lastly, the hardware prototype is implemented on a development board; continued work involves custom PCB miniaturization to allow for full integration into a pen form factor for real-world use. Possible application areas are secure banking, IoT access control, healthcare authentication, and electronic contract signing. These lines show the extent of scalability of the system into a deployable, privacy-enhancing, and user-friendly authentication platform.

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